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Do Real Estate Values Boost Corporate Borrowing? Evidence from Contract-Level Data

Contents

1	Big picture	3
2	Data and measurement	3
2.1	Commercial real estate transactions data	3
2.2	Matched RCA–Compustat sample	4
2.3	Debt contract data	4
2.4	Real estate value	4
2.5	Systematic risk exposures	4
2.6	Investment, controls, and firm characteristics	5
3	Model and empirical specifications	5
3.1	Theory: secured versus unsecured borrowing	5
3.2	Recovery rates and observed secured-debt pricing	5
3.3	Investment specification	6
3.4	Instrument for real estate values	6
3.5	Debt issuance specification	6
3.6	Risk exposure interaction	6
3.7	Real estate transaction dynamics	7
4	Identification and empirical design	7
4.1	Sources of variation	7
4.2	What the investment regression identifies	7
4.3	What the debt contract regressions identify	8
4.4	Addressing alternative explanations	8
5	Tables and figures	8
5.1	Figure 1 and Figure 2: empirical fact and theoretical mechanism	8
5.2	Tables 1–3 and Figure 3: sample construction and debt categories	9
5.3	Figures 4–6 and Table 4: real estate price indices and risk exposures	9
5.4	Table 5: real estate values and investment	10
5.5	Table 6 and Figure 7: debt issuance by contract and collateral type	11
5.6	Table 7 and Figure 8: risk exposure and debt issuance	11
5.7	Figure 9 and Table 8: debt holdings and determinants of risk exposure	12
5.8	Figure 10 and Table 9: property portfolio dynamics	13
5.9	Tables 10–12: covenants, flexibility, and Compustat universe	13
5.10	Figure 11 and Tables 13–15: debt levels, debt changes, and tradable sector	14
6	Short summary	14

7	Conclusion	15
7.1	Core conclusion	15
7.2	Economic intuition	15
7.3	Takeaway	15

1 Big picture

- **Question.** Do increases in the value of firms' real estate assets boost corporate borrowing through a collateral channel, and if so, what type of debt do firms issue?
- **Core motivation.** A large literature argues that pledgeable collateral relaxes borrowing constraints. The canonical prediction is that increases in collateral values should raise secured borrowing and investment.
- **Main contribution.** The paper uses contract-level debt data and transaction-level commercial real estate data to show that the collateral channel is more subtle than the standard view suggests.
- **Key new mechanism.** Real estate is not only collateral; it is an asset with its own systematic risk exposure. If real estate values are highly exposed to aggregate or sectoral downturns, then lenders may demand worse secured-debt terms, making unsecured debt more attractive.
- **Empirical setting.** The authors build a matched sample of Compustat firms and Real Capital Analytics commercial real estate transactions over 2000–2017. They combine these data with detailed corporate debt contract data from Capital IQ and LPC DealScan.
- **Main investment result.** Higher real estate values are associated with higher corporate investment, consistent with a collateral channel at the investment margin.
- **Main financing result.** The debt response is not concentrated in secured borrowing. It is concentrated in unsecured bonds and notes, especially senior unsecured bonds and notes.
- **Risk result.** Firms whose real estate assets have higher systematic risk relative to their overall asset portfolio tilt more strongly toward unsecured debt when real estate values increase.
- **Why this matters.** Pledgeability depends not only on collateral value but also on collateral quality in bad states. An asset can be valuable yet poor collateral if it is likely to lose value exactly when the borrower defaults.
- **Takeaway.** Real estate values do boost corporate activity, but the financing response often bypasses secured credit. The reason is that firms and lenders account for the systematic risk of the collateral itself.

2 Data and measurement

2.1 Commercial real estate transactions data

- The authors use Real Capital Analytics (RCA) transaction-level data on commercial real estate acquisitions, dispositions, and refinancing transactions in the United States.
- RCA records property characteristics such as location, property type, buyer, seller, transaction date, transaction price, and property use.
- The RCA sample includes properties worth at least \$1 million and covers transactions between 2000 and 2017.
- The authors validate and clean RCA records using additional sources and link properties to public Compustat firms using firm names and ownership information.
- The data identify both where firms are headquartered and where their real estate assets are located, which is essential because headquarters locations alone do not reveal the locations of corporate real estate portfolios.

Interpretation: This transaction-level dataset is central because it allows the authors to observe the market value and geographic location of corporate real estate holdings, rather than relying only on accounting book values or headquarters-based proxies.

2.2 Matched RCA–Compustat sample

- The matched sample contains 9,219 firm-year observations from 654 publicly listed Compustat firms with real estate holdings.
- Firms in finance, insurance, real estate, construction, and mining are excluded because their real estate holdings may serve different purposes or their debt structure may be institutionally different.
- The matched sample resembles the larger Compustat universe of real estate holding firms along industry composition, but it has richer data on property-level holdings.
- Figure 3 shows that sample firms’ headquarters and property locations are widely distributed across the United States.

Interpretation: The sample is not simply a sample of real estate firms. It is a sample of ordinary operating firms whose balance sheets include real estate assets that may serve as collateral.

2.3 Debt contract data

- Debt contract information comes from Capital IQ and LPC DealScan.
- The paper distinguishes revolving credit, term loans, senior secured bonds and notes, senior unsecured bonds and notes, junior unsecured bonds and notes, and other debt categories.
- Secured debt is further classified by collateral type: real estate, property/plant/equipment including real estate, and “all assets.”
- The dependent variables in the debt issuance regressions are debt issuance amounts scaled by lagged fixed assets.

Interpretation: Contract-level data allow the paper to test whether the collateral channel operates through the specific debt instruments that are actually secured by real estate, rather than through aggregate leverage alone.

2.4 Real estate value

- The main real estate value measure is the market value of firms’ corporate real estate holdings, scaled by lagged fixed assets.
- The authors infer market values by combining RCA transaction prices, property locations, and CBSA-level commercial real estate price indices.
- The true market value measure is denoted $REValue_{i,t-1}^{True}$.
- The value measure varies across firms because firms hold different types and locations of real estate assets.

Interpretation: Scaling by lagged fixed assets makes the real estate value comparable across firms and ties the measure to firms’ borrowing and investment capacity.

2.5 Systematic risk exposures

- The authors estimate the systematic risk exposure of each firm’s real estate portfolio, $\beta_{i,t}^{RE}$.
- They also estimate the systematic risk exposure of the firm’s overall asset portfolio, $\beta_{i,t}^{Firm}$.
- The key relative risk measure is

$$\beta_{i,t}^{Diff} = \beta_{i,t}^{RE} - \beta_{i,t}^{Firm}. \quad (1)$$

- If $\beta_{i,t}^{Diff} > 0$, the firm's real estate assets are systematically riskier than the firm's overall balance sheet.
- If $\beta_{i,t}^{Diff} < 0$, the firm's real estate assets are relatively safe collateral.

Interpretation: This relative-risk measure is the paper's key innovation. The question is not merely whether real estate is valuable, but whether it is valuable in the states of the world in which lenders care most.

2.6 Investment, controls, and firm characteristics

- Investment outcomes include total investment, three-year mean investment, and industry-adjusted investment.
- Controls include cash flow, Tobin's Q , firm size, and establishment-portfolio growth.
- Fixed effects include firm fixed effects and CBSA-by-time fixed effects.
- Standard errors are clustered by CBSA-year.

Interpretation: The fixed effects and controls are designed to separate the effect of a firm's own real estate value from local macroeconomic and industry trends.

3 Model and empirical specifications

3.1 Theory: secured versus unsecured borrowing

The model's core idea is that firms choose between secured and unsecured debt by comparing the costs of borrowing from lenders whose recoveries depend differently on asset values in default states.

- Secured lenders price debt using expected collateral recovery.
- If collateral values are likely to fall in distress states, secured lenders require larger haircuts or higher spreads.
- Unsecured lenders have claims on the broader asset pool and may be less exposed to the risk of a specific collateral asset.
- Firms with risky collateral may therefore issue unsecured debt even when collateral values rise.

Interpretation: The model modifies the standard collateral-channel intuition. Higher collateral value can raise debt capacity, but high collateral risk can shift the composition of borrowing away from secured debt.

3.2 Recovery rates and observed secured-debt pricing

The model compares expected recovery rates for secured and unsecured lenders. For a lender L , expected recovery depends on the expected collateral value in bankruptcy states, haircuts, and the borrower's choice of debt type:

$$RR_L = E[\text{recovery of lender } L \mid \text{borrower default}]. \quad (2)$$

- Secured claims backed by systematically risky collateral can have lower expected recovery in bad states.
- Observed recovery rates can reflect both lender protection and borrower selection into secured versus unsecured debt.
- The theory predicts that firms with relatively risky real estate should avoid secured borrowing backed by that real estate.

Interpretation: Secured debt is not always safer in economic terms. A claim backed by an asset that collapses in distress can be riskier than a broader unsecured claim.

3.3 Investment specification

The investment regression is

$$Investment_{i,t} = \alpha_i + \beta_{l,t} + \gamma REValue_{i,t-1}^{True} + \theta Controls_{i,t-1} + \epsilon_{i,t}. \quad (14)$$

- α_i are firm fixed effects.
- $\beta_{l,t}$ are headquarters-location-by-time fixed effects.
- $REValue_{i,t-1}^{True}$ is instrumented.
- The coefficient γ measures how real estate values affect firm investment.

Interpretation: This tests the conventional collateral-channel prediction: if real estate appreciates, firms should invest more because their balance sheets become stronger.

3.4 Instrument for real estate values

The first-stage instrument is based on local land supply elasticity interacted with the national mortgage rate:

$$REIndex_{l,t} = \alpha_l + \beta_t + \gamma Elasticity_l \times Mortgage_t + \varepsilon_{l,t}. \quad (15)$$

- Locations with less elastic land supply experience larger real estate price responses to changes in national mortgage rates.
- The instrument generates plausibly exogenous variation in local real estate prices.
- The estimated real estate price index is then used to instrument firms' real estate values.

Interpretation: The IV strategy follows the idea that national credit conditions have heterogeneous effects across cities depending on local supply constraints.

3.5 Debt issuance specification

Debt issuance is estimated using

$$Debt_{i,t} = \alpha_i + \beta_{l,t} + \gamma REValue_{i,t-1}^{True} + \theta Controls_{i,t-1} + \epsilon_{i,t}. \quad (16)$$

- $Debt_{i,t}$ is alternative issuance of debt by type and collateral category.
- The specification is estimated separately for total debt, secured debt, and unsecured debt categories.
- The same instrumented real estate value measure is used.

Interpretation: This regression asks where the borrowing response occurs. The standard collateral channel predicts more real-estate-secured debt; the paper finds that the response is instead concentrated in unsecured bonds and notes.

3.6 Risk exposure interaction

To test the role of collateral risk, the authors estimate variants with interactions:

$$Debt_{i,t} = \alpha_i + \beta_{l,t} + \gamma_1 REValue_{i,t-1}^{True} + \gamma_2 Positive \beta_{i,t}^{Diff} + \gamma_3 REValue_{i,t-1}^{True} \times Positive \beta_{i,t}^{Diff} + \theta Controls_{i,t-1} + \epsilon_{i,t}. \quad (17)$$

Interpretation: If risky real estate collateral drives the preference for unsecured debt, the positive unsecured-debt response should be stronger when β^{Diff} is positive.

3.7 Real estate transaction dynamics

The paper models the choice to purchase, sell-and-leaseback, outright sell, or retain a property using a multinomial logit:

$$\ln \left(\frac{Pr(Y_{i,j,l,t} = k)}{Pr(Y_{i,j,l,t} = 4)} \right) = \gamma_k \beta_{i,l,t}^{RE} + \theta_k Controls_{i,j,l,t-1} + \alpha_j + \alpha_l + \alpha_t. \quad (19)$$

- $Y = 1$ denotes purchase.
- $Y = 2$ denotes sale-and-leaseback.
- $Y = 3$ denotes outright sale.
- $Y = 4$ denotes retain.

Interpretation: This examines whether firms actively select, retain, or dispose of real estate assets based on systematic risk exposure.

4 Identification and empirical design

4.1 Sources of variation

- Variation in local real estate prices comes from CBSA-level real estate price indices.
- Instrumental variation comes from land supply elasticity interacted with national mortgage rates.
- Variation in systematic risk exposure comes from the locations and property types of firms' real estate holdings.
- Variation in debt outcomes comes from contract-level data that distinguish between secured and unsecured instruments.

Interpretation: The research design combines local real estate shocks with firm-specific real estate portfolios and contract-level debt choices.

4.2 What the investment regression identifies

The investment regression identifies whether higher real estate values relax financing constraints enough to raise real activity.

- Firm fixed effects absorb persistent differences across firms.
- CBSA-by-time fixed effects absorb local economic shocks at headquarters locations.
- Controls absorb time-varying firm fundamentals such as cash flow, Q , and size.

Interpretation: A positive effect on investment confirms that real estate values matter for firm financing capacity, but it does not yet reveal the type of borrowing used.

4.3 What the debt contract regressions identify

The debt regressions identify how real estate value shocks translate into debt issuance by contract type.

- Total debt tests the broad collateral channel.
- Real-estate-secured debt tests the narrow collateral channel.
- Unsecured bonds and notes test the paper’s alternative prediction.
- Interactions with β^{Diff} test whether debt composition depends on the riskiness of collateral.

Interpretation: The key empirical surprise is that real estate values boost investment and total borrowing while leaving real-estate-secured borrowing mostly unchanged.

4.4 Addressing alternative explanations

- Debt covenants could prevent firms from pledging assets; the paper controls for negative pledge covenants and tangible net worth covenants.
- Financial flexibility could explain avoidance of secured debt; the paper controls for pre-existing secured debt and operating concentration.
- Matched-sample peculiarities could drive results; the paper replicates tests in the larger Compustat universe.
- The issuance-versus-level distinction could matter; the paper repeats tests using changes in debt holdings.
- Sector-specific demand shocks could matter; the paper repeats tests for firms in the tradable sector.

Interpretation: These robustness exercises are designed to show that the central finding is not an artifact of covenants, operational flexibility, sample selection, or local demand shocks.

5 Tables and figures

5.1 Figure 1 and Figure 2: empirical fact and theoretical mechanism

Read:

- Figure 1 shows that the real capital stock of US public firms rises over time, while the share of firms raising mortgage debt remains small.
- Figure 1 also shows that commercial real estate values rise substantially in the sample period.
- Figure 2 illustrates the model’s comparative statics: unsecured debt, real estate values, and systematic risk exposure are jointly connected.

Economic message: The paper starts with a puzzle: firms hold valuable real estate, but they do not predominantly finance themselves with mortgage debt. The model explains this using collateral risk.

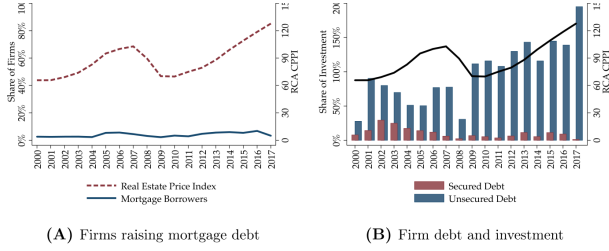


Figure 1: Real estate holdings and borrowing by US public firms. Panel A depicts the share of US public firms raising mortgage debt, alongside the Real Capital Analytics Commercial Property Price Index for the US (RCA CPTI, 2006=100). Panel B depicts real estate- and PBE-secured debt versus all unsecured debt used as a share of firm investment, alongside the RCA CPTI. Data sources: Capital IQ, Compustat, DealScan, and RCA.

Figure 1: real estate holdings and borrowing

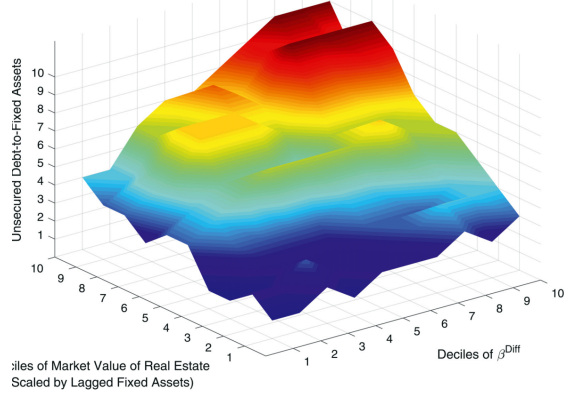


Figure 2: Model mechanism. This figure shows cross-correlations between unsecured debt, real estate values, and systematic risk exposures. Unsecured debt is measured by the residual unsecured debt-to-fixed assets ratio, purged of common determinants of leverage. The market value of corporate real estate, defined in Section 3.4, is also scaled by lagged fixed assets. β^{Diff} is the difference between a firm's

Figure 2: model mechanism

5.2 Tables 1–3 and Figure 3: sample construction and debt categories

Read:

- Table 1 presents the model's expected recovery rates for secured and unsecured lenders.
- Table 2 shows that the RCA–Compustat matched sample resembles the broader Compustat universe of real estate holding firms by industry composition.
- Figure 3 shows broad geographic coverage of headquarters and real estate holdings.
- Table 3 shows that bonds and notes are the largest share of debt; senior unsecured bonds and notes are particularly important.

Economic message: These visuals justify the empirical design: the sample is broad, the real estate holdings are geographically dispersed, and the debt data are detailed enough to distinguish secured from unsecured instruments.

Lender			Expected Recovery Rate
L_A^U	$RR_A^U = 1 - \frac{1}{U_A} \left[\beta^{firm} \left[\bar{F} - \sum \phi(k_A) \right] \right]$	(Observed)	
L_B^S	$RR_B^S = 1 - \frac{1}{S_B} \left[\beta^{firm} \left[\bar{F} - \sum \phi(k_B) \right] \right]$	(Observed)	
L_A^S	$RR_A^S = 1 - \frac{1}{S_A} \left[\beta^{firm} \left[\bar{F} - \sum \phi(k_A) \right] \right]$	(Unobserved)	
L_B^U	$RR_B^U = 1 - \frac{1}{U_B} \left[\beta^{firm} \left[\bar{F} - \sum \phi(k_B) \right] \right]$	(Unobserved)	

This table presents expected recovery rates, $RR \in \{RR_A^U, RR_B^S, RR_A^S, RR_B^U\}$, of lenders, $L \in \{L_A^S, L_B^U, L_A^U, L_B^S\}$.

above inequality. In other words, the data will show that

(a) Table 1: recovery rates.

Industry composition by number of firm-year observations.				
Industry	Sample		Compustat Universe	
	Frequency	Share	Frequency	Share
Consumer Non-durables	837	0.09	2613	0.09
Consumer Durables	268	0.03	1273	0.04
Manufacturing	1543	0.17	5532	0.19
Energy	60	0.01	155	0.01
Chemicals	242	0.03	1389	0.05
Business Equipment	1702	0.18	4855	0.17
Telecommunications	257	0.03	934	0.03
Shops	2386	0.26	4995	0.18
Healthcare	845	0.09	2743	0.10
Other	1079	0.12	3,931	0.14
Total	9219	1.00	28,420	1.00

This table presents the distribution of industries in our sample as compared to the industry composition of the Compustat universe of real estate-holding firms over the 2000–2017 period.

(b) Table 2: industry composition.

Debt contract	Mean share of total debt			
	Sample	Compustat Universe	Colla et al. (2013)	Security status
Commercial paper	0.02	0.01	0.01	Unsecured
Drawn credit lines	0.13	0.18	0.22	Secured
Term loans	0.16	0.22	0.21	Secured
Bonds and notes	0.58	0.49	0.48	Mixed
Senior secured	0.13	0.20	N/A	Secured
Senior unsecured	0.73	0.63	N/A	Unsecured
Junior	0.13	0.17	N/A	Unsecured
Capital leases	0.06	0.06	0.05	N/A
Other	0.05	0.04	0.03	N/A

This table presents the breakdown of total debt for the average firm in our sample into six comprehensive and mutually exclusive categories following Colla et al. (2013). The table also presents the comparison for the average Compustat firm, as well as the breakdown reported in Colla et al. The final column reports the security status of each type of debt as either secured or unsecured. The data used to produce this table are obtained from Capital IQ for the 2000–2017 period.

(c) Table 3: debt breakdown.

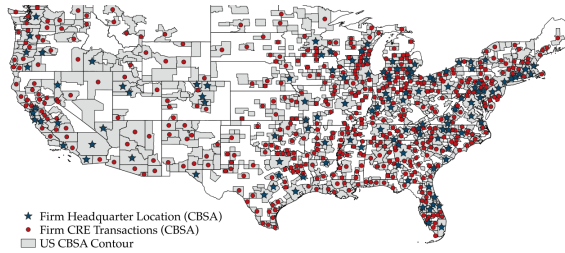


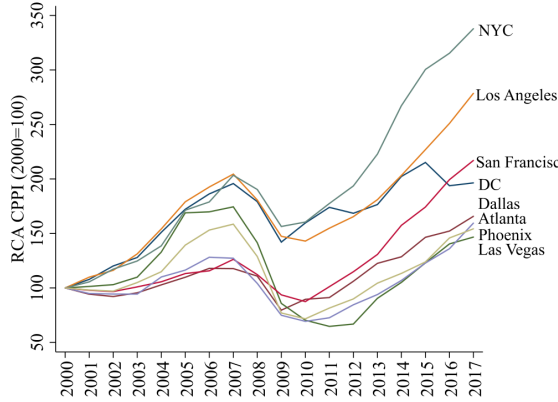
Figure 3: Headquarters and real estate holdings locations of RCA–Compustat matched sample firms. The figure shows the geographical distribution across US

5.3 Figures 4–6 and Table 4: real estate price indices and risk exposures

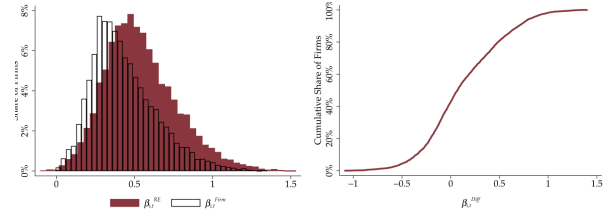
Read:

- Figure 4 shows that CBSA-level commercial real estate prices vary significantly across cities.
- Figure 5 shows that real estate systematic risk exposures differ from firm-level systematic risk exposures.
- The distribution of β^{Diff} implies that for many firms, real estate is systematically riskier than the firm's overall asset portfolio.
- Figure 6 shows that unsecured debt is more expensive than secured debt for risky-real-estate firms, while the pattern differs for safe-real-estate firms.
- Table 4 reports the main variables used in the empirical analysis.

Economic message: The data support the model's core premise: real estate collateral has heterogeneous systematic risk, and this risk is priced in secured versus unsecured debt costs.



(a) Figure 4: CBSA real estate price indices.



(b) Figure 5: systematic risk exposure distributions.

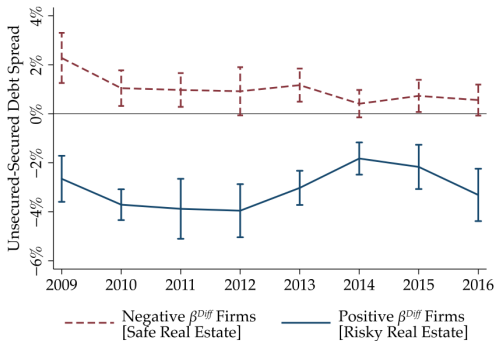


Figure 6: unsecured-secured debt spread

Variable	Sample			Compustat		
	Mean	Median	N	Mean	Median	N
Investment						
Total Investment	0.25	0.19	8,893	0.23	0.17	27,738
3-Year Mean Investment	0.29	0.20	7,570	0.25	0.18	22,231
Industry-Adjusted Investment	-0.30	-0.14	8,892	-0.33	-0.17	27,726
Total Debt Issuance						
All Long-Term Debt	0.40	0.00	8,944	0.37	0.00	27,825
Issuance of Revolving Credit Secured by						
Real Estate	0.00	0.00	8,944	0.00	0.00	27,825
Other PPGE	0.01	0.00	8,944	0.01	0.00	27,825
All Assets	0.13	0.00	8,944	0.11	0.00	27,825
Issuance of Term Loans Secured by						
Real Estate	0.00	0.00	8,944	0.00	0.00	27,825
Other PPGE	0.00	0.00	8,944	0.01	0.00	27,825
All Assets	0.06	0.00	8,944	0.08	0.00	27,825
Issuance of Bonds and Notes						
Senior Secured	0.01	0.00	8,944	0.01	0.00	27,825
Senior Unsecured	0.18	0.00	8,944	0.14	0.00	27,825
Subordinated	0.01	0.00	8,944	0.01	0.00	27,825
Characteristics						
RE Value	0.60	0.31	8,601	0.62	0.35	25,796
RE Value ^{True}	0.74	0.45	8,601	N/A	N/A	N/A
Cash Flow	0.52	0.40	8,910	0.44	0.34	27,814
Q	2.06	1.61	7,848	1.84	1.47	24,027
Size	7.83	7.81	9,073	6.96	6.87	28,420
Establishment-Portfolio Growth	0.03	0.04	9,219	0.03	0.04	28,420

This table presents descriptive statistics for the main variables used in our empirical analyses over the 2000-2017 period. The investment measures are from Compustat. Data on the issuance of bonds and notes in the categories of Senior Secured, Senior Unsecured, and Subordinated bonds and notes are from Capital IQ. Data on the issuance of revolving credit secured by Real Estate, Other PPGE, and All Assets, as well as data on the issuance of term loans secured by Real Estate, Other PPGE, and All Assets are from LPC DealScan. Total debt issuance, denoted All Long-Term Debt, is the sum of all issuances under the contract and collateral types stated above. All investment and debt issuance variables are scaled by lagged fixed assets. RE Value is calculated following CST. RE Value^{True} is calculated based on firms' true geographical real estate holdings, as described in Section 3.4. Cash Flow is income before extraordinary items (IB) plus depreciation and amortization (DP), scaled by lagged fixed assets. Q is the market-to-book ratio. Size is the natural logarithm of total assets (AT). Establishment-Portfolio Growth is annual per capita personal income growth, weighted by a given firm's exposure to different CBSAs based on the firm's

Table 4: descriptive statistics

5.4 Table 5: real estate values and investment

Read:

- Table 5 estimates the investment regression using instrumented real estate values.
- The coefficient on REValue^{True} is positive and statistically significant for total investment, three-year mean investment, and industry-adjusted investment.
- The estimates imply that rising real estate values increase corporate investment.
- This confirms the investment side of the collateral channel.

Economic message: Real estate appreciation has real effects on firm investment. The remaining question is how firms finance that response.

Investment as a function of corporate real estate values.

	Total investment 1	3-Year Mean investment 2	Industry-adjusted investment 3
<i>RE Value^{True}</i>	0.077*** (0.016)	0.106*** (0.024)	0.072** (0.032)
<i>Cash Flow</i>	0.018*** (0.006)	0.023*** (0.007)	0.011 (0.010)
<i>Q</i>	0.037*** (0.005)	0.043*** (0.006)	0.046*** (0.009)
<i>Size</i>	0.008 (0.010)	-0.035** (0.015)	-0.001 (0.019)
<i>Establishment-Portfolio Growth</i>	0.257 (0.452)	0.201 (0.678)	0.494 (0.968)
Instrumented <i>RE Value^{True}</i>	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes
CBSA×Time Fixed Effects	Yes	Yes	Yes
Observations	5,869	4,894	5,868
R-squared	0.51	0.64	0.42

This table reports output from Eq. (14). The dependent variable is *Total Investment* in the first specification, *3-Year Mean Investment* in the second specification, and *Industry-Adjusted Investment* in the final specification. All investment measures are based on Compustat item *CAPX* and scaled by lagged fixed assets. *RE Value^{True}* is the market value of corporate real estate assets, scaled by lagged fixed assets, measured at time $t - 1$. *RE Value^{True}* is estimated based on instrumented CBSA-level real estate price indices obtained from Eq. (15). *Cash Flow* is income before extraordinary items plus depreciation, scaled by lagged fixed assets, measured at time $t - 1$. *Q* is the market-to-book ratio, measured at time $t - 1$. *Size* is the logarithm of total assets, measured at time t . *Establishment-Portfolio Growth* is annual per capita personal income growth, weighted by a given firm's exposure to different CBSAs based on the firm's operations in each CBSA, measured at time $t - 1$. Firm and CBSA×time fixed effects are included as indicated. All regressions are estimated over the 2000–2017 period. Robust standard errors, reported in parentheses, are clustered by CBSA×year. Statistical significance is indicated as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

channel narrative. The coefficients in column 1 are on corporate real estate. This result would support

5.5 Table 6 and Figure 7: debt issuance by contract and collateral type

Read:

- Table 6 decomposes debt issuance by contract and collateral type.
- The total-debt coefficient on $REValue^{True}$ is positive and significant.
- Real-estate-secured revolving credit, term loans, and senior secured bonds do not respond meaningfully to real estate values.
- Senior unsecured bonds and notes respond positively and significantly.
- Figure 7 visualizes the result: senior unsecured bond issuance rises with real estate values, while senior secured bond issuance is nearly flat.

Economic message: The central result contradicts the narrow collateral channel. Higher real estate values raise borrowing, but mostly through unsecured bonds and notes rather than real-estate-secured debt.

5.6 Table 7 and Figure 8: risk exposure and debt issuance

Read:

- Table 7 interacts real estate value with an indicator for positive β^{Diff} .
- Total debt and unsecured debt responses are stronger when firms' real estate assets are systematically riskier than their overall asset portfolios.
- Figure 8 shows that for firms with positive β^{Diff} , the marginal effect on unsecured debt issuance is high, while the effect on secured debt issuance is close to zero.

Debt issuance by contract and collateral type as a function of corporate real estate values.

	Total debt				Secured debt				Unsecured Debt	
	Revolving credit				Term loans				Bonds and Notes	
	RE	PPBE	"All Assets"		RE	PPBE	"All Assets"		Senior	Junior
1	2	3	4	5	6	7	8	9	10	
RE Value ⁷⁰⁰	0.300***	0.001	-0.016**	0.077	-0.000	0.001	0.068**	0.000	0.151***	0.017***
Cash Flow	0.049	-0.000	0.001	-0.011	0.000	0.000	0.020	-0.002*	0.040**	0.000
Q	-0.033*	-0.000	-0.002	-0.003	0.000*	-0.000	-0.001	0.001	-0.027***	-0.001
Size	0.314***	0.000	-0.005	0.082**	-0.000*	0.002	0.070**	-0.003	0.169***	-0.001
Establishment-Portfolio Growth	(0.074)	(0.001)	(0.005)	(0.034)	(0.000)	(0.002)	(0.030)	(0.004)	(0.042)	(0.003)
Observations	5,890	5,890	5,890	5,890	5,890	5,890	5,890	5,890	5,890	5,890
R-squared	0.35	0.40	0.38	0.32	0.29	0.26	0.29	0.25	0.40	0.32

This table reports output from Eq. (16). The specification reported in column 1 refers to the issuance of Total Debt. The specifications in columns 2 through 8 refer to the issuance of secured debt, by contract and collateral type. Contract types include Revolving Credit, Term Loans, and Senior Secured Bonds and Notes. Under Revolving Credit and Term Loans, collateral types include real estate (RE), property, plant, and equipment including real estate (PPBE), and all assets (All Assets). The specifications in columns 9 and 10 refer to the issuance of Unsecured Bonds and Notes by claim priority, i.e., Senior or subordinated (Junior). In each case, issuance is calculated as the sum of all issuances of a given type of debt contract, with a given collateral or seniority, in each firm-year, scaled by lagged fixed assets. The data for issuance of Bonds and Notes are based on security-level issuance information reported on Capital IQ. The data for issuance of Revolving Credit and Term Loans by collateral type are obtained from LPC DealScan. RE Value⁷⁰⁰ is the market value of corporate real estate holdings, measured at time $t-1$. RE Value⁷⁰⁰ is estimated based on instrumented CREA-level real estate price indices obtained from Eq. (15). Cash Flow is income before extraordinary items plus depreciation, scaled by lagged fixed assets, measured at time $t-1$. Q is the market-to-book ratio, measured at time $t-1$. Size is the logarithm of total assets, measured at time t . Establishment-Portfolio Growth is annual per capita personal income growth, weighted by a given firm's exposure to different CSAs based on the firm's operations in each CSA, measured at time $t-1$. Firm and CSA-time fixed effects are included as indicated. All regressions are estimated over the 2000-2017 period. Robust standard errors, reported in parentheses, are clustered by



Figure 8. Marginal effects of real estate value shocks on corporate debt issuance. The figure depicts margin of contract in a given firm-year across different levels of Market Value of Real Estate (RE Value⁷⁰⁰). Amount of Debt Issued is scaled by lagged fixed assets. The estimated margins are derived from the model estimates in columns 7 and 8

- The evidence connects debt composition directly to real estate systematic risk.

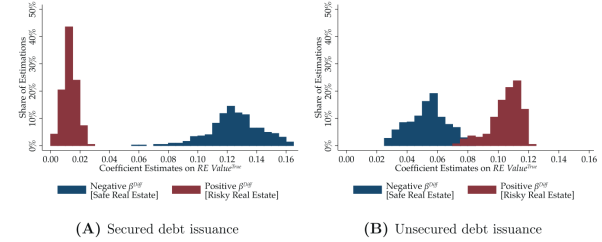
Economic message: Firms with risky collateral prefer unsecured debt. This is the paper's most direct evidence for the theoretical mechanism.

Table 7
Debt issuance by contract and collateral type as a function of corporate real estate values and real estate risk exposures.

	Total debt				Secured debt				Unsecured Debt	
	Revolving credit				Term Loans				Bonds and Notes	
	RE	PPBE	"All Assets"		RE	PPBE	"All Assets"		Senior	Junior
1	2	3	4	5	6	7	8	9	10	
RE Value ⁷⁰⁰ × Positive β^{Diff}	0.437***	0.000	-0.003	0.263***	0.000	-0.001	0.071**	-0.002	0.100***	0.009
RE Value ⁷⁰⁰	(0.124)	(0.001)	(0.002)	(0.101)	0.000	(0.001)	(0.033)	(0.004)	(0.040)	(0.007)
Cash Flow	0.068	0.002	-0.008	-0.100	0.000	-0.001	0.020	0.002	0.148*	0.005
Positive β^{Diff}	(0.206)	(0.001)	(0.006)	(0.144)	(0.001)	(0.001)	(0.043)	(0.008)	(0.088)	(0.010)
Q	-0.371**	0.001	0.003	-0.250***	0.000	0.002	-0.079	0.009	-0.015	-0.001
Size	(0.189)	(0.001)	(0.004)	(0.127)	(0.001)	(0.003)	(0.058)	(0.008)	(0.087)	(0.012)
Establishment-Portfolio Growth	0.112	0.000	-0.002	-0.011	0.000	0.000	0.104	-0.007	0.029	-0.001
Observations	(0.161)	(0.000)	(0.002)	(0.025)	(0.000)	(0.000)	(0.090)	(0.079)	(0.001)	
R-squared	-0.045	0.000	0.001	-0.013	0.000	0.000	-0.014	0.001	-0.022	0.001
Instrumented RE Value ⁷⁰⁰	(0.047)	0.000	(0.001)	(0.026)	0.000	0.000	(0.022)	(0.002)	(0.022)	(0.002)
Firm Fixed Effects	0.473***	0.001	0.002	0.118	-0.001*	0.001	0.124**	-0.010	0.241***	-0.002
CSA-Time Fixed Effects	(0.169)	(0.001)	(0.003)	(0.101)	(0.001)	(0.002)	(0.052)	(0.008)	(0.089)	(0.005)
Observations	-2.043	0.031	-0.616	-0.030	-0.047	-1.180	0.195	-0.073	-0.269*	
R-squared	(0.386)	(0.072)	(0.193)	(2.464)	(0.021)	(0.054)	(0.826)	(0.202)	(1.593)	(0.153)

This table reports output from Eq. (11). The specification reported in column 1 refers to the issuance of Total Debt. The specifications in columns 2 through 8 refer to the issuance of secured debt, by contract and collateral type. Contract types include Revolving Credit, Term Loans, and Senior Secured Bonds and Notes. Under Revolving Credit and Term Loans, collateral types include real estate (RE), property, plant, and equipment including real estate (PPBE), and all assets (All Assets). The specifications in columns 9 and 10 refer to the issuance of Unsecured Bonds and Notes by claim priority, i.e., Senior or subordinated (Junior). In each case, issuance is calculated as the sum of all issuances of a given type of debt contract, with a given collateral or seniority, in each firm-year, scaled by lagged fixed assets. The data for issuance of Bonds and Notes are based on security-level issuance information reported on Capital IQ. The data for issuance of Revolving Credit and Term Loans by collateral type are obtained from LPC DealScan. RE Value⁷⁰⁰ is the market value of corporate real estate holdings, measured at time $t-1$. RE Value⁷⁰⁰ is estimated based on instrumented CREA-level real estate price indices obtained from Eq. 10. Positive β^{Diff} is the difference between the firm's exposure to different CSAs based on the firm's operations in each CSA, measured at time $t-1$. Firm and CSA-time fixed effects are included as indicated. All regressions are estimated over the 2000-2017 period. Robust standard errors, reported in parentheses, are clustered by

Figure 8. Marginal effects of real estate value shocks on corporate debt issuance. Panel A (B) depicts the histograms of coefficient estimates on the market value of real estate. The x-axis is 'Coefficient Estimates on RE Value⁷⁰⁰'. The y-axis is 'Share of Estimations' from 0% to 50%. Two series are shown: Negative β^{Diff} (Sale Real Estate) and Positive β^{Diff} (Risky Real Estate). Panel A shows a distribution skewed towards positive estimates, while Panel B shows a distribution skewed towards negative estimates.



5.7 Figure 9 and Table 8: debt holdings and determinants of risk exposure

Read:

- Figure 9 shows predicted secured and unsecured debt holdings across deciles of real estate value and β^{Diff} .
- Secured debt holdings are relatively flat with respect to real estate values.
- Unsecured debt holdings increase strongly when real estate values rise, especially for firms with high β^{Diff} .
- Table 8 shows that real estate systematic risk is related to geographic concentration, property-type concentration, major market share, and redeployability.

Economic message: The debt-level evidence reinforces the issuance results. Firms' balance-sheet debt structure tilts toward unsecured debt when real estate is risky collateral.

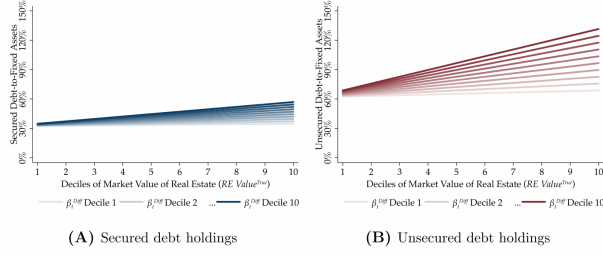


Fig. 9: Debt holdings as a function of the market value of firms' real estate. Panel (A) depicts the predicted values of firms' secured (unsecured) debt holdings, across the deciles of the $RE\text{Value}^m$ distribution for firms in different deciles of the β^{Diff} distribution. The estimated margins are derived from β^{Diff} .

Figure 9: debt holdings

	Count of real estate transactions				Value of real estate transactions			
	1	2	3	4	5	6	7	8
Geographical HHI	0.018*** (0.003)	0.019*** (0.004)	0.018*** (0.004)	0.020*** (0.004)	0.068*** (0.006)	0.068*** (0.006)	0.069*** (0.005)	0.069*** (0.006)
Property Type HHI	0.093*** (0.005)	0.092*** (0.005)	0.098*** (0.005)	0.094*** (0.006)	0.012*** (0.005)	0.008*** (0.003)	0.011*** (0.004)	0.010*** (0.005)
Major Market Share	-0.018*** (0.008)	-0.017*** (0.009)	-0.018*** (0.009)	-0.018*** (0.007)	-0.041*** (0.005)	-0.040*** (0.006)	-0.044*** (0.009)	-0.042*** (0.009)
High Redevelop Share	-0.044*** (0.005)	-0.045*** (0.005)	-0.045*** (0.004)	-0.044*** (0.004)	-0.011*** (0.002)	-0.016*** (0.002)	-0.015*** (0.002)	-0.015*** (0.002)
Controls								
Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects								
State	Yes	No	Yes	No	Yes	No	Yes	No
Industry	No	Yes	No	No	No	Yes	No	No
State × Industry	No	No	No	Yes	No	No	No	Yes
Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,274	1,274	1,274	1,274	1,274	1,274	1,274	1,274
R-squared	0.70	0.70	0.70	0.70	0.69	0.69	0.69	0.69

This table reports output from Eq. (17). The dependent variable is β_{it}^{Diff} . β_{it}^{Diff} denotes systematic risk exposure, defined as the difference between β_{it}^{RE} , obtained from Eq. (12), and β_{it}^{Diff} , obtained from Eq. (13). Geographical HHI is the Herfindahl-Hirschman Index of a firm's real estate holdings across the 160 geographical markets defined by RCA, where holdings are measured in counts (values) in columns 1 through 4 (5 through 8). Property Type HHI is the Herfindahl-Hirschman Index of a firm's real estate holdings across the 8 property types defined by RCA, where holdings are measured in counts (values) in columns 1 through 4 (5 through 8). Major Market Share is the share of a firm's real estate holdings located in the 27 major geographical markets for which RCA computes commercial real estate price indices, where holdings are measured in counts (values) in columns 1 through 4 (5 through 8). High Redevelop Share is the share of firm's real estate holdings that belong to development site, office, industrial, or retail property types, where holdings are measured in counts (values) in columns 1 through 4 (5 through 8). Robust standard errors, reported in parentheses, are clustered by CBSA-year. Statistical significance is indicated as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8: determinants of systematic risk exposure

5.8 Figure 10 and Table 9: property portfolio dynamics

Read:

- Figure 10 illustrates geographic dispersion of holdings for firms with low and high β^{Diff} .
- Firms with high risk exposure tend to hold concentrated assets in important markets.
- Table 9 estimates transaction dynamics in a multinomial logit framework.
- High β^{RE} is associated with different probabilities of purchasing, sale-and-leaseback, outright sale, or retention.

Economic message: Firms' real estate portfolios are not random. They actively acquire, retain, or dispose of properties in ways related to systematic risk exposure.

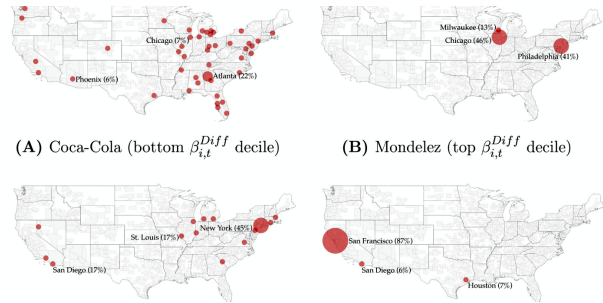


Figure 10: geographical dispersion

Table 9
Systematic risk exposures and transaction dynamics.

	Purchase 1	Sale-and-leaseback 2	Outright sale 3
β^{RE}	0.110 (0.201)	-1.436*** (0.302)	-2.266*** (0.331)
Size	0.361*** (0.021)	0.457*** (0.119)	-0.557*** (0.082)
Price	-0.016 (0.038)	0.459*** (0.132)	-0.036 (0.095)
Age	-0.337*** (0.108)	-0.965*** (0.175)	-0.791*** (0.123)
Firm Fixed Effects	Yes		
CBSA Fixed Effects	Yes		
Time Fixed Effects	Yes		
Observations	38,013		
Pseudo R-squared	0.34		

This table presents output from Eq. (19). The reported coefficients are estimated via a multinomial logistic regression in which the outcomes refer to whether a given property in a given year is purchased, sold-and-leased back, sold outright, relative to retained (baseline outcome). β^{RE} is as

Table 9: transaction dynamics

5.9 Tables 10–12: covenants, flexibility, and Compustat universe

Read:

- Table 10 controls for debt covenants, including negative pledge and tangible net worth covenants.

- Table 11 controls for financial and operational flexibility using pre-existing secured debt and operating concentration.
- Table 12 replicates the main debt-issuance tests in the larger Compustat universe.
- Across these tests, the core pattern remains: unsecured bonds and notes respond to real estate values, while real-estate-secured debt does not.

[illegible][illegible]

5.10 Figure 11 and Tables 13–15: debt levels, debt changes, and tradable sector
Read:

- Figure 11 transforms the debt-holdings evidence into the balance between unsecured and secured debt.
- Table 13 uses changes in debt holdings rather than issuance.
- Table 14 repeats the changes-in-debt tests in the Compustat universe.
- Table 15 repeats the main debt issuance tests for firms operating in the tradable sector.
- The results remain consistent with the central conclusion: real estate value shocks are reflected in unsecured debt rather than real-estate-secured debt.

Economic message: The robustness evidence shows that the paper’s findings survive alternative dependent variables, broader samples, and concerns about local demand factors.

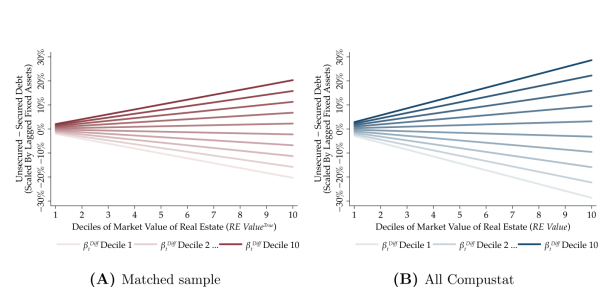


Fig. 11. Balance of debt holdings as a function of the market value of firms' real estate. The figure depicts the predicted values for the balance of unsecured and secured debt holdings across the deciles of the $REValue^{TNC}$ distribution for firms in different deciles of the β^{DWI} distribution, relative to those for firms in the middle of that distribution. The estimated margins are derived from Eq. (16). The lower (higher) deciles of the β^{DWI} distributions include the

6 Short summary

- The paper studies whether increases in corporate real estate values boost borrowing and investment.
- It combines transaction-level commercial real estate data, Compustat firm data, and contract-level debt data.

Changes of debt holdings as a function of corporate real estate values for the Compustat universe.

	Total debt		Secured debt				Unsecured debt	
	1	Revolving credit		Term loans		Bonds and notes	Bonds and notes	
		PP&E	"All Assets"	PP&E	"All Assets"			
RE value	0.664*** (0.101)	0.213* (0.117)	0.019 (0.092)	0.233 (0.192)	0.127 (0.097)	0.077 (0.158)	0.249*** (0.057)	
Cash flow	0.022 (0.034)	-0.011 (0.021)	0.007 (0.016)	-0.012 (0.030)	-0.016 (0.017)	-0.031 (0.037)	0.000 (0.023)	
Q	0.329*** (0.044)	0.155** (0.066)	0.037 (0.028)	-0.009 (0.064)	0.127*** (0.033)	0.014 (0.038)	0.083*** (0.026)	
Size	0.743*** (0.074)	0.152 (0.132)	0.046 (0.048)	0.213** (0.108)	0.172*** (0.063)	0.230** (0.109)	0.343*** (0.044)	
Instrumented RE Value	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
CBSA-Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Observations	15,238	599	2,846	457	1,824	2,668	8,774	
r-squared	0.23	0.61	0.43	0.63	0.51	0.40	0.28	

This table reports output from Eq. (11) for the changes of total, secured, and unsecured debt holdings in the Compustat universe of real estate-holding firms. Column 1 presents the results for total long-term debt. Columns 2 through 6 present the results for secured debt contracts (by contract and collateral type). Column 7 presents the results for unsecured debt contracts. Under secured debt contracts, we consider the following contract types. Columns 2 and 3 present estimates for changes in Revolving Credit. Columns 4 and 5 present estimates for changes in Term Loans. Column 6 presents estimates for changes in Secured Bonds and Notes. Column 7 presents estimates for changes in Unsecured Bonds and Notes. Under Revolving Credit and Term Loans, we consider changes of debt holdings for each of those contract categories by the type of collateral pledged; namely, property, plant, and equipment including real estate (PP&E) in columns 2 and 4, and "All Assets" in columns 3 and 5. Under Secured Bonds and Notes we focus on senior facilities outstanding. Under Unsecured Bonds and Notes we focus on the sum of senior and junior facilities outstanding. Changes in all debt contracts stated above are computed as the annual rate of growth in the amount of debt outstanding under a given contract type, with a given collateral type or level of seniority. Data for the amounts of debt outstanding are obtained from DealScan (for Revolving Credit and Term Loans by collateral type) and from Capital IQ (for Secured and Unsecured Bonds and Notes), respectively. RE Value is the market value of corporate real estate holdings, measured at time $t-1$. RE Value is estimated based on instrumented CBSA-level real estate price indices obtained from Eq. 10. Cash flow is income before extraordinary items plus depreciation, scaled by

Debt issuance by contract and collateral type as a function of corporate real estate values for firms in the tradable sector.

	Total debt		Secured debt				Unsecured Debt	
	1	Revolving credit		Term loans		Bonds and notes	Bonds and Notes	
		RE	PP&E	"All Assets"	RE	PP&E	"All Assets"	
RE Value ^{time}	0.284** (0.114)	0.001 (0.001)	-0.012* (0.007)	0.070 (0.065)	-0.000 (0.000)	0.002 (0.003)	0.069** (0.034)	0.000 (0.004)
Cash Flow	0.044 (0.034)	-0.000 (0.000)	0.001 (0.001)	-0.011 (0.010)	0.000 (0.000)	0.000 (0.000)	0.021 (0.019)	-0.002 (0.001)
Q	-0.039* (0.020)	-0.000 (0.000)	-0.002 (0.002)	-0.001 (0.010)	0.000 (0.000)	-0.000 (0.001)	-0.001 (0.011)	-0.000 (0.001)
Size	0.315*** (0.078)	0.000 (0.001)	-0.006 (0.005)	0.077** (0.036)	-0.000 (0.000)	0.002 (0.002)	0.070** (0.033)	-0.004 (0.004)
Establishment-portfolio growth	-0.306 (2.334)	-0.010 (0.035)	0.002 (0.160)	0.640 (1.573)	-0.017 (0.011)	-0.028 (0.076)	0.112 (0.907)	0.341** (0.133)
Instrumented RE Value ^{time}	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
CBSA-Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5116	5116	5116	5116	5116	5116	5116	5116
R-squared	0.35	0.39	0.37	0.32	0.26	0.26	0.30	0.26

This table reports output from Eq. (16) for the sample firms operating in the tradable sector. The specification reported in column 1 refers to the issuance of Total Debt. The specifications in columns 2 through 8 refer to the issuance of secured debt, by contract and collateral type. Contract types include Revolving Credit, Term Loans, and Senior Secured Bonds and Notes. Under Revolving Credit and Term Loans, collateral types include real estate (RE), property, plant, and equipment including real estate (PP&E), and all assets (All Assets). The specifications in columns 9 and 10 refer to the issuance of Unsecured Bonds and Notes by claim priority, i.e., Senior or subordinated (Junior). In each case, issuance is calculated as the sum of all issuances of a given type of debt contract, with a given collateral or seniority, in each firm-year, scaled by lagged fixed assets. The data for issuance of Bonds and Notes are based on security-level issuance information reported on Capital IQ. The data for issuance of Revolving Credit and Term Loans by collateral type are obtained from LPC DealScan. RE Value^{time} is the market value of corporate real estate holdings, measured at time $t-1$. RE Value^{time} is estimated based on instrumented CBSA-level real estate price indices obtained from Eq. (15). Cash flow is income before extraordinary items plus depreciation, scaled by lagged fixed assets, measured at time $t-1$. Q is the market-to-book ratio, measured at time $t-1$. Size is the logarithm of total assets, measured at time t . Establishment-Portfolio Growth is

- Real estate values are instrumented using land supply elasticity interacted with national mortgage rates.
- Higher real estate values raise investment, validating the broad collateral-channel intuition.
- However, higher real estate values do not primarily increase real-estate-secured borrowing.
- Instead, firms increase unsecured bonds and notes.
- The reason is collateral risk: real estate with high systematic risk is poor collateral in bad states.
- Firms with positive β^{Diff} respond to higher real estate values by issuing more unsecured debt.
- Secured debt issuance is largely insensitive to real estate value shocks.
- Robustness checks rule out covenants, flexibility concerns, sample selection, changes-versus-levels, and local demand as explanations.
- The paper's key lesson is that collateral value and collateral risk jointly determine corporate debt structure.

7 Conclusion

7.1 Core conclusion

Campello, Connolly, Kankanhalli, and Steiner show that real estate values do boost corporate investment and borrowing, but not in the simple way implied by a narrow collateral channel. Firms do not mainly respond by issuing more real-estate-secured debt. Instead, they often issue more unsecured bonds and notes.

7.2 Economic intuition

Collateral is valuable only if it remains valuable when lenders need it. If a firm's real estate portfolio is exposed to systematic shocks, then the collateral may lose value exactly when the firm defaults. In that case, secured borrowing against real estate becomes less attractive, and the firm may rationally prefer unsecured debt backed by the broader asset pool.

7.3 Takeaway

The paper adds a risk dimension to the collateral channel. Increases in collateral value can relax financing constraints and raise investment, but the type of debt firms choose depends on the systematic risk of the collateral. A complete theory of corporate borrowing must therefore consider not only how much collateral firms have, but also how risky that collateral is in bad states of the world.